Nguyễn Đại Đồng

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dong-nguyen-hcmut Noname-29-lnin

Education

VNU - Ho Chi Minh City University of Technology *Oct 2022 – Present*

Bachelor of Science in Electronics Engineering

- **GPA:** 3.1/4.0 (This link)
- **Project 1:** Design and Implementation Viterbi Decoder Algorithm on FPGA.
- **Project 2:** Hardware Accelerator for Mean-Based Large-Scale Sorting using PIM Architecture.

Research Interest

Computer Architecture, RISC-V Processors, Hardware Accelerato, Digital IC Design.

Technical Skill

- Programming Language & HDL:
  - Proficient: C, C++, SystemVerilog, Verilog.
  - Scripting & Documentation: LaTeX, Bash.
  - Tools: Vim, GIT, Makefile.
- Development Environments:
  - Operating Systems: Linux (Ubuntu, CentOS), Windows.
  - FPGA Tools: Intel Quartus, QuestaSim, ModelSim, Verilator, Virtuoso, Xcelium.
- Hardware Skills:
  - Digital Design: RTL development, Verification.
  - PCB Design: KiCad, Alitium (Basic).
  - Lab Equipment: Oscilloscope, Logic Analyzers.

Projects

Design and Implementation of a Power-Efficient Viterbi Encoding and Decoding Architecture on FPGA: From Theory to Practical Application *Noname-29-lnin/Viterbi_decoder*

- This project involves the design and implementation of a **Viterbi decoder** for convolutional codes with parameters $(3, 1, 2)$ (constraint length $K = 3$, code rate $1/2$) on an FPGA. The Viterbi algorithm is optimized for error correction in noisy communication channels, leveraging trellis-based path metrics and survivor path selection. The decoder is structured using key functional blocks, including the *Branch Metric Unit (BMU)*, *Add-Compare-Select Unit (ACSU)*, *Path Metric Unit (PMU)*, and *Survivor Path Memory Unit (SPMU)*.

- To test the decoder, additional modules such as a *Parallel-In-Serial-Out (PISO)* register, *Serial-In-Parallel-Out (SIPO)* register, and *LCD IP core* (for hardware-based LCD control) are integrated. These components enable practical verification of the Viterbi decoder's functionality. The final implementation is deployed on a *DE2 development kit*, where the design is synthesized and programmed onto the FPGA for real-world validation.
- Tools & Lauguage:
  - Simulation: Verilator, GTKwave.
  - Synthesis: Intel Quartus Prime.
  - Language: C, SystemVerilog.

Hardware Accelerator for Mean-Based Large-Scale Sorting using PIM Architecture *Noname-29-lnin/Sort_FPGA*

- Designed and implemented the hardware architecture for a high-throughput sorting framework, based on the research *"A General Framework for Sorting Large Data Sets Using Independent Subarrays of Approximately Equal Length"*.
- The RTL implementation targets performance optimization for large-scale datasets (from 2^{18} to 2^{20}+ elements), supporting multiple data types.
- Engineered the core processing unit to handle independent subarrays, optimizing for either area (serial processing) or throughput (parallel processing) to balance resource utilization and sorting speed on the FPGA.
- **Tools & Technologies:**
  - **Simulation:** Verilator, QuestaSim, GTKWave.
  - **Synthesis & Implementation:** Intel Quartus Prime.
  - **Languages:** SystemVerilog (RTL Design), C++, Python.